

Dok. 261 — Static Geometry and the SI Projection: Why Natural Units Require No Fractal Corrections

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.1 Setting

The 2019 SI reform fixed four constants ($c, \hbar, e, k_B$) to exact values and thereby turned most dimensionful quantities into mere unit definitions. What remains as *genuine* empirical input are the dimensionless constants – above all the fine-structure constant α , whose value is still measured after the reform (μ_0, ε_0 and α remained measured quantities). FFGFT derives these dimensionless inputs as well, from the single parameter $\xi = \frac{4}{3} \times 10^{-4}$.

This document collects the conversion machinery compactly (Tables 1–3) and then establishes one central point: *as long as one stays in natural units and pure ratios, the description is static and relative and requires no fractal corrections*. The fractal corrections appear only upon projection to absolute SI values.

.2 Compactified time: static and complete — the arrow makes it dynamic

This section sets the frame above all the others: whether a quantity appears static or dynamic depends solely on the *status of the time direction* – not on the number of dimensions.

The full FFGFT geometry is the 4-torus $T^4 = T^3 \times S^1$, in which time is a *compact* direction – the time-winding S^1 (Dok. 230). In this full view time is a closed loop, on equal footing with the three spatial windings; there is no flowing “now” and no arrow. The description is purely relational: winding numbers and dimensionless ratios. This is *layer 1* of Dok. 241 – the level on which FFGFT makes its fundamental statements.

On this layer everything appears static, yet nothing is lost. The theory’s fractal recursion is not absent here but *frozen*: a ratio such as $m_\mu/m_e \approx 206.768$ is the end product of a recursion that has come to rest in this number – just as the golden ratio φ does not stand *beside* the Fibonacci sequence but *is* its limit (Dok. 241). What is missing on layer 1 is not the fractality but its visible motion.

Static is not incomplete. In the compact T^4 (time as a winding) the description carries all invariant content – as frozen fixed points of fractal recursions. The transition to dynamics adds no content; it merely sets the recursion into visible motion.

Dynamics arises through *decompactification*: when the time-winding S^1 is unrolled into the open arrow t – geometrically the step $T^4 \rightarrow T^3$ that “loses the time-winding” (Dok. 230) – three spatial directions plus one directed time remain: the observed 3+1 world. What singles out this one direction as an arrow is the fundamental ξ asymmetry that “enables all evolution processes” (Dok. 078). Only now does the recursion run visibly: processes unfold, phase accumulates, and the *embedding correction* becomes explicit – the “price” of which Dok. 185 speaks.

With this, everything that follows falls into one picture. The bridge from the static layer 1 to the dynamic 3+1 world – *layer 2*, the SI translation (Dok. 241) – is carried by exactly $c, \hbar, \varepsilon_0$. That is why these constants become real only here (Section .7),

and why G and α too are not static fundamental quantities but dynamic couplings (Section .8). The only static fundamental quantity remains ξ ; everything else is either a frozen ξ fixed point (layer 1) or its visibly moving projection (layer 2).

A clarification so that "complete" is not misread: it means *contains all invariant content*, not *the dynamics is unreal*. For the observer embedded in the arrow, the time-flow is operationally real. The clean statement is: the compact T^4 picture is complete in content; the dynamics is its real but content-neutral internal perspective.

Finally, the arc back to retrocausality closes. On the T^4 with the time-winding intact, a closed timelike curve is merely a winding of the time-loop, and entanglement appears as a *topological connection* (Dok. 230) – static geometry. Only after decompactification into 3+1 arrow-time does the same geometry look like backward causation or nonlocality. The apparent exoticism of the retrocausal is thus an artifact of the dynamic internal view, not of the geometry.

.3 The constants fixed by the SI reform

Table 1: The constants fixed by the 2019 SI reform (the "machinery"). They carry no physics, only scale, and become 1 in natural units.

Constant	Value (exact since 2019)	defines	natural units
c	299 792 458 m/s	metre	= 1
$\hbar$	$1.054 571 817 \dots \times 10^{-34}$ J s	kg	= 1
e	$1.602 176 634 \times 10^{-19}$ C	ampere	charge unit
k_B	$1.380 649 \times 10^{-23}$ J/K	kelvin	= 1

.4 Quantity conversion in natural units

.5 The dimensionless inputs and their derivation from ξ

.6 Static geometry: why natural units are correction-free

As long as physics is expressed in natural units and pure ratios, its description is *static* and *relational*: only dimensionless ratios occur, no absolute anchors and no scale dependence. At this level the FFGFT quantities are exact geometric invariants of ξ – and no fractal correction is required.

Principle. In natural units FFGFT is purely relational. The dimensionless structural relations – mass ratios, mixing angles, the Koide quantity – are static geometric invariants of ξ . They contain no fractal corrections.

Table 2: Quantity conversion in natural units ($\hbar = c = k_B = 1$): all dimensionful quantities become powers of energy.

Quantity	SI dimension	$[\hbar = c = k_B = 1]$	back to SI
Energy	J	E	base
Mass	kg	E	$\times S_{T0}$ (via c^{-2})
Momentum	kg m/s	E	$\times c^{-1}$
Length	m	E^{-1}	$\ell[\text{fm}] = 197.327/E[\text{MeV}]$
Time	s	E^{-1}	$t[\text{s}] = 6.582 \times 10^{-22}/E[\text{MeV}]$
Temperature	K	E	$T[\text{K}] = E[\text{eV}]/(8.617 \times 10^{-5})$
Charge (Gauss)	C	dimensionless	$e = \sqrt{\alpha} \approx 0.0854$

The conversion to SI runs through the *single* mass scale $S_{T0} = 1 \text{ MeV}/c^2 = 1.782662 \times 10^{-30} \text{ kg}$ (Dok. 013/014); the length factor is $\hbar c/S_{T0} = 1.973 \times 10^{-13} \text{ m}$, i.e. the 197 MeV fm of the $\hbar c = 1$ column expressed in T0 language.

In T0 units the charge unit is chosen ($e = \sqrt{4\pi\epsilon_0\hbar c}$) so that the *written* value is $\alpha = 1$ (Dok. 044) – a convention, not a physical intervention. The physical invariant $\alpha \approx 1/137$ is reconstructed from ξ (Table 3).

Table 3: The surviving dimensionless inputs of physics and their derivation from the single parameter $\xi = \frac{4}{3} \times 10^{-4}$. All relations are sourced.

Invariant	Standard physics	FFGFT relation (from ξ)	Source
ξ	—	$= \frac{4}{3} \times 10^{-4} = 4/30000$	Dok. 013
α (SI)	1/137.036	$\xi(E_0/1 \text{ MeV})^2$ [†]	Dok. 011
Lepton masses	measured	$m = r \xi^p \nu$	Dok. 116, 006
exponents		$(p_e, p_\mu, p_\tau) = (\frac{3}{2}, 1, \frac{2}{3})$	Dok. 116
ratios		$(r_e, r_\mu, r_\tau) = (\frac{4}{3}, \frac{16}{5}, \frac{25}{9})$	Dok. 116
Koide $\mathcal{Q}$	0.66666	0.667742 (exactly $\frac{2}{3}$ only at ξ_{Koide})	Dok. 116, 258, 259
$\sin^2 \theta_W$	0.2312	$\frac{1 - \sqrt{1 - 4\alpha_W}}{4}$, $\alpha_W = \sqrt{\xi}$ [‡]	Dok. 041
θ_{13}	8.57°	$\arcsin(\xi^{1/3})$ [‡]	Dok. 041
G	measured	$\frac{\xi^2}{4m_e} K_{\text{frak}}$, $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$	Dok. 012, 133

[†] $\alpha = \xi(E_0/1 \text{ MeV})^2$ is the *leading order*, with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398 \text{ MeV}$ (bare $\sqrt{m_e m_\mu} = 7.348 \text{ MeV}$, Doc. 190, R72). The agreement with the measured value is a property of the anchor E_0 , which provides the only connection to the SI, not an independent prediction.

[‡] With $\alpha_W = \sqrt{\xi} = 0.01155$ the $\sin^2 \theta_W$ formula gives 0.0058 and misses the measured value 0.2312; $\arcsin(\xi^{1/3})$ gives 2.93°, not 8.57° (Doc. 041).

CODATA precision is reached through the fractal correction product $\prod_{n=1}^{137} (1 + \delta_n \xi (4/3)^{n-1})$. These corrections are **not free assumptions**: $K_{\text{frak}} = 1 - 100\xi$ is uniquely fixed by the scale flow from the Planck to the electroweak scale ($D_f^{\text{space}} = 3 - \xi$, Dok. 133, “ D_f is uniquely determined – not freely choosable”), and the rational ladder ratios r_i are α -geometric invariants (Dok. 258), structurally inaccessible to any after-the-fact fitting (non-closure theorem, Dok. 193).

The clearest example is the Koide quantity: from the exponents $(p_e, p_\mu, p_\tau) = (\frac{3}{2}, 1, \frac{2}{3})$ and the ratios $(r_e, r_\mu, r_\tau) = (\frac{4}{3}, \frac{16}{5}, \frac{25}{9})$ with $\xi = 4/30000$ – with no additional parameter and no fractal correction (Dok. 116) – one obtains $Q = 0.667742$, i.e. +0.16 % above $\frac{2}{3}$ ($\Delta_K = 3Q - 2 = +0.0032$; Dok. 258, 259); exactly $\frac{2}{3}$ is obtained only at $\xi_{\text{Koide}} = 1.3718 \times 10^{-4}$. Likewise the ladder ratios r_i and the exponents p_i are exact functions of ξ alone; the angle formulas of Table 3 are also pure ξ expressions, but their values for $\sin^2 \theta_W$ and θ_{13} miss the measured values (footnote ‡). None of these relations needs an absolute scale, an E_0 in MeV, or a conversion factor.

Where, then, do the fractal corrections arise? Only at the *SI projection* – where an absolute scale is fixed. $K_{\text{frak}} = 1 - 100\xi$ is not a correction to the geometry but the cumulative effect of the scale flow from the Planck to the electroweak scale (Dok. 133): a property of mapping the geometry onto an absolute energy ruler. Likewise the fractal correction product $\prod_n (1 + \delta_n \xi (4/3)^{n-1})$ appears only when the leading relation $\alpha = \xi(E_0/1 \text{ MeV})^2$ is brought to CODATA precision in absolute SI values.

An important delimitation: these projection factors are nonetheless *not* free parameters. K_{frak} is uniquely fixed by the scale flow (Dok. 133), the rational ratios are α -geometric invariants (Dok. 258), and the non-closure theorem (Dok. 193) blocks any after-the-fact adjustment. They are derived projection factors, not assumed fits.

Conclusion. The fractal corrections are the price of the absolute projection, not a defect of the geometry. As long as one stays in natural units and ratios, the description is static, relative and correction-free; the corrections appear exclusively at the transition to absolute SI values – and even there they are uniquely determined by ξ and the scale structure.

.7 Dynamic reality of the constants: $c, \hbar, \varepsilon_0$ in the time-flow

The previous section showed that the static ξ -geometry is correction-free. The remaining question is *when* the constants $c, \hbar$ and ε_0 acquire physical reality at all. The answer follows from the same split, viewed from the other side: these constants are not properties of the geometry at rest but the conversion rates it adopts as soon as it is embedded in a real, time-extended process.

FFGFT distinguishes two times here (Dok. 078): the intrinsic, characteristic scale $\tilde{T}$ of the duality $\tilde{T} \cdot m = 1$, which governs discrete system properties, and the overarching continuous time-flow (the "arrow time" t) in which systems exist and processes unfold. The duality concerns the intrinsic scale, not the time-flow itself. Only in the time-flow – when a timeless identity is physically *realized* – does what Dok. 185 calls the *embedding price* arise: realizing a static mathematical fact as a process costs real time. $c, \hbar$ and ε_0 are the exchange rates of this embedding price.

Embedding price. A timeless ξ -relation is a static fact. Its physical realization as a process in arrow-time costs time – and c , $\hbar$, ε_0 are exactly the rates that quantify this price (Dok. 185).

In detail:

c is not a fundamental law of nature but a dynamic ratio L/T (Dok. 134). Statically, energy and mass are the same number ($E = m$, $c = 1$, length $\equiv$ time); c does not appear at all. As soon as something propagates, c becomes the traversal rate of the T^4 geometry – the full form $E = mc^2$, the dispersion $E^2 = (pc)^2 + (mc^2)^2$, the light cone. c is real as the rate at which the geometry is traversed in time.

$\hbar$ is the exchange rate between geometric winding and accumulated real time. Statically, phase is a dimensionless winding number on the torus ($\hbar = 1$). In the time-flow the phase $e^{-iEt/\hbar}$ accumulates; $\hbar$ is the rate at which the geometric winding becomes visible as oscillation, interference and decay. In the field form $T_{\text{field}} \cdot E_{\text{field}} = 1$ (Dok. 020) this is the transition from the intrinsic scale to observable dynamics.

ε_0 is the vacuum's response to time-varying fields. Statically it is absorbed into the charge-unit convention $\alpha = 1$. Dynamically it becomes real via $c = 1/\sqrt{\varepsilon_0\mu_0}$: ε_0 and μ_0 resonate dually (vacuum as a resonant medium, Dok. 043/044), and their product is the propagation of light in time. ε_0 is real as the impedance of the vacuum to the temporal field flow.

The consequence for particle physics is a sharp dividing line:

Demarcation. *Dimensionless, static* quantities (mass ratios, mixing angles, $\sin^2 \theta_W$, the Koide quantity Q) live entirely in the ξ -geometry – without c , $\hbar$, ε_0 and without fractal correction. *Absolute, time-extended* quantities (lifetimes, decay rates, cross sections, oscillation frequencies, propagators) exist only in the time-flow, require c , $\hbar$, ε_0 to be real, and thereby inherit the SI projection (S_{T0} , K_{frak}).

This is concretely testable: a lifetime *ratio* (two processes, $\hbar$ cancels) is derivable cleanly from ξ alone (cf. Dok. 160); an *absolute* lifetime carries the dynamic constants and the projection factors. The demarcation thus predicts which quantities FFGFT delivers sharply and correction-free and which must pass through the projection.

A closing ontological clarification: "real" here means *operationally real in the dynamic regime*, not *ontologically fundamental*. The geometry (ξ , T^4 , $\tilde{T} \cdot m = 1$) is primary; c , $\hbar$, ε_0 are the real conversion rates the timeless web of ratios adopts once it becomes process-like. They are the SI shadows of embedding the static geometry into the time-flow.

.8 Consequence: G and α are not static fundamental constants

Exactly the same demarcation applies to the two quantities traditionally regarded as fundamental constants: the gravitational constant G and the fine-structure constant α . Both are *coupling constants* – they quantify the strength of an interaction. And a coupling strength is the clearest case of a dynamic quantity: it describes a process and it *runs* with scale.

α . Structurally α is not a free input but derived from ξ : $\alpha = \xi(E_0/1 \text{ MeV})^2$ (Dok. 011, cf. Table 3); that this relation reproduces the measured value is a property of the anchor E_0 , which provides the connection to the SI, and does not count as an independent prediction. Operationally, however, $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ – built from exactly the embedding constants $\epsilon_0, \hbar, c$ – and it *runs* with energy ($\alpha(Q^2)$). This running is nothing other than the scale flow that also generates the fractal correction product $\prod_n(1 + \delta_n \xi(4/3)^{n-1})$. Thus α is neither fundamental (it follows from ξ) nor static (it lives in the scale/time-flow).

G . In natural units $G = 1$ (Dok. 012). The SI value follows from ξ : $G = \frac{\xi^2}{4m_e} K_{\text{frak}}$ – thereby containing the projection factor $K_{\text{frak}} = 1 - 100\xi$, which is itself a scale-flow effect (Dok. 133). The equivalent Planck form $G = \ell_P^2 c^3/\hbar$ (Dok. 012/127) shows it even more directly: G is expressed through $c^3/\hbar$, i.e. through the dynamic constants themselves. G is thus doubly projected – via K_{frak} and via $c, \hbar$ – and measures the strength of the gravitational interaction, a process. G too is neither fundamental nor static.

Conclusion. The only static fundamental quantity is ξ . G and α are not fundamental, static constants but *dynamic couplings*: derived from ξ , dressed with the embedding constants $c, \hbar, \epsilon_0$ and clothed by scale-flow factors (K_{frak} , the fractal product). Like $c, \hbar, \epsilon_0$, they become real only in the time-flow. The apparent fundamentality of G and α is a consequence of the measurement convention, not a foundation.

.9 The demarcation from the Lagrangian viewpoint

The static/dynamic split of this document is not a reading of Tables 1–3 but the architecture of the Lagrangian extension itself. The tables only summarize what the Lagrangian already carries.

Two densities as a necessity. Dok. 095 shows that FFGFT requires two complementary Lagrangian densities – not by style but from the scale hierarchy. The simplified

$$\mathcal{L}_{\text{T0}} = \varepsilon(\partial\delta E)^2, \quad \varepsilon = \xi E^2$$

carries pure ξ structure (layer 1); the extended Standard-Model density with T0 corrections carries the dynamics (layer 2). Both are, as Dok. 095 stresses, “mathematical descriptions ... not nature itself”. The demarcation is thus a corollary of the two densities, not an interpretive overlay.

The cut runs term by term. The full decomposition (Dok. 067) separates derivative from mass terms:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu T \partial_\nu T - V(T) \right], \quad \mathcal{L}_{\text{gauge}} = \sqrt{-g} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right),$$

$$\mathcal{L}_\psi = \sqrt{-g} \Omega^4(T) (i\bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi).$$

The *derivative terms* ($\partial_\mu, \sqrt{-g}$ with the metric, $-\frac{1}{4}F^2$) carry the dynamics – and, in dimensional form, c (via the metric) and $\hbar$ (via the kinetic normalization). The

mass/potential terms ($V(T)$, $m\bar{\psi}\psi$) are static. The Euler-Lagrange step that turns these into equations of motion is the decompactification into arrow-time of Section .2.

α and G in the Lagrangian. The gauge coupling sits in the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ (Dok. 019) – so α is structurally a derivative (dynamic) term, not a static parameter. And gravity is not inserted but *emergent*: Dok. 049 shows it arises automatically from $\mathcal{L} = \varepsilon(\partial\delta m)^2$, and Dok. 180 derives G in step 4 of a self-consistency chain directly from $\xi - \bar{\xi}$, L_0 and G are not postulated but follow from the self-consistency of the Lagrangian.

Field-theoretic origin. The demarcation of Sections .2, .7 and .8 is a consequence of the Lagrangian extension: the two densities (Dok. 095) are layers 1 and 2; the cut between derivative and mass terms (Dok. 067) is the static/-dynamic cut; α lives in the covariant derivative (Dok. 019); G emerges from ξ (Dok. 049/180). The tables are the SI summary, not the source of proof.